

Active v0.4.0, frozen. Deployed by explicit decision after two gated remediation attempts failed; trades 2021-2023-vintage development AUC for current-generator accuracy.

Task: real-versus-AI classification with a label, AI-positive score, CPU web/API/batch interface and model-influence explanation. Public-data results only; organizer data, baseline and hidden scores were not supplied.

Model, data and operating point

Frozen generic **CLIP ViT-L/14** image tower plus our trained **2-layer MLP head** (768->128->1, ReLU, dropout 0.3), replacing v0.3.0's linear head. Training: **8,000 CIfAKE + 6,239 GenImage + 1,183 AIDA-train (17 current generators) + 2,122 CommunityForensics-train (~20 generators)**. 60/40 holdouts of both new sources fixed before training and never used for fitting.

Inference: EXIF orientation, official 224px CLIP centre crop, L2-normalized embedding, MLP head, temperature scaling. Temperature **0.226417**; AI threshold **0.838809**. Threshold uses internal GenImage/CIfAKE validation only; neither holdout was used to fit it.

Best-epoch selection (epoch 60/60) used internal validation AUC only; no holdout or external data influenced model/epoch selection.

Runnable interface and attribution

Shared CPU CLI/API; single image or batch with one loaded model. JPEG/PNG/WebP uploads up to **25 MiB / 40 MP**. EXIF is separate; C2PA presence is a heuristic substrng scan, not signature verification.

Tower **608,352,029 bytes** (reused unchanged from v0.3.0, parity verified) + head **398,527 bytes**, SHA-256 verified. **CPU-only** model runtime: torch, without clip/torchvision/ftfy/regex.

Measured CPU components (shared CLIP tower): predict **~0.38s/image**, explanation **~5.5s**, robustness **~5.1s**. Fresh-clone timing not yet re-measured for this release.

Why this model exists: the gate failed twice, deployed by explicit decision

Two independent 2025-2026-generator evaluations on the previous release (v0.3.0): AI Detect Arena (17 generators) macro AUC **0.602**, accuracy **55.2%**, real-photo FPR **68.3%**. A model flagging two-thirds of real photos as AI is not usable. **Two gated remediation attempts both failed** on external-dev regression (linear+AIDA: 0.771->0.652 AUC; MLP+AIDA+CommunityForensics: 0.771->0.637 AUC), yet the MLP attempt's own untouched holdouts reached usable accuracy. Deployed by explicit user decision favouring current-generator accuracy over the old development benchmark - both facts disclosed together, not one substituted for the other.

Measured result: this release on its own untouched holdouts

Holdout (never used in training)	Accuracy	Real->real	AI->AI	AUC
AI Detect Arena (17 current generators, n=755)	85.2%	92.7%	78.5%	0.948
CommunityForensics-Eval (~20 generators, n=1374)	71.8%	97.8%	54.4%	0.936

Real-photo recognition is strong on both (92.7%/97.8%). AI recall is decent on AIDA (78.5%), weaker on CommunityForensics (54.4%, which mixes older GAN generators with current diffusion models). Aggregate accuracy looks better than AI-recall alone because real photos dominate correctness in both sets. CommunityForensics-Eval is CC BY-NC-SA 4.0 (Park et al., CVPR 2025).

Cost: the 2021-2023-vintage development benchmark this release trades away

External dev (GLIDE-family generators)	v0.3.0	v0.4.0
Mean AUC, as distributed	0.771	0.637
Mean AUC, format matched	0.789	0.658

No GLIDE/DALLE or COCO reserved-set number is claimed for v0.4.0: that evaluation ran for v0.2.0/v0.3.0 only, and re-running it a third time was not judged worth spending that evidence on given the deployment decision already accepts this tradeoff. B-Free (third-party, CVPR 2025) scored 0.970/0.945 on the same 2021-2023-vintage development images without fitting - not our result.

Limitations

Explanation audit not re-run for this architecture: the 17/40-localised, p=0.29/0.82 findings in prior reports are from v0.3.0's linear head. Input-gradient explanation works for the MLP head (verified live) but has not had its own 40-image audit. **Module G robustness** and the **private user-image veto check** have likewise not been re-run for v0.4.0 as of this report. No generator attribution, caption consistency or C2PA signature verification. No organizer hidden-test score is available or claimed.

Sources: github.com/AI-Detect-Arena/benchmark-dataset; huggingface.co/datasets/OwensLab/CommunityForensics-Eval (Park et al., CVPR 2025); github.com/WisconsinAIVision/UniversalFakeDetect; github.com/openai/CLIP; github.com/GenImage-Dataset/GenImage; CIfAKE (Bird & Lotfi). Code and full experiments: github.com/sibtainmunshi/Signal_Scope.
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